Sublinear separators and expansion: the exact value of Dvořák's exponent b_ε

Machine-generated note (see provenance box)

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Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The construction and proof were found by the language model `gpt-5.6-sol` in a single autonomous pass on 2026-08-31, as part of a large sweep over open problems catalogued from arXiv (catalog id `2001.09679__00`; original writeup in `attacks/2001.09679__00/output.md`). (2) An independent adversarial referee agent (`claude-fable-5`) re-derived every step, checked the source paper, and brute-forced the finite structural claims on a concrete instance; its verdict was MINOR_GAPS, with two presentational gaps (the definition of b_ε, and a one-sentence padding remark). Its report is reproduced verbatim in Appendix B. (3) On 2026-09-09 the writeup was rewritten into the present note by `claude-fable-5.1`: the two gaps were repaired (Section 5 and Proposition 6.2), a reference for the expander input was added, and the text was reorganised, with full proofs moved to Appendix A. No mathematical content beyond these repairs was added. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

Abstract

For a hereditary class $\mathcal{G}$ of graphs let $s_{\mathcal{G}}(n)$ be the largest size of a minimum balanced separator over graphs of $\mathcal{G}$ with at most n vertices, and let $\nabla_{\mathcal{G}}(r)$ be the largest density of an r-shallow minor of a graph in $\mathcal{G}$. Dvořák (2020) defined b_ε as the supremum of the exponents b such that every hereditary class with $s_{\mathcal{G}}(n) = \Theta(n^{1-\varepsilon})$ has $\nabla_{\mathcal{G}}(r) = \Omega(r^b)$, proved $\frac{1}{2\varepsilon} - 1 \le b_\varepsilon \le \frac{1}{2\varepsilon} - \frac{1}{2}$ for $0 < \varepsilon < \frac{1}{2}$, and asked which bound is tight. We construct, for every $0 < \varepsilon < \frac{1}{2}$, a monotone class $\mathcal{C}_\varepsilon$ with $s_{\mathcal{C}_\varepsilon}(n) = \Theta(n^{1-\varepsilon})$ and $\nabla_{\mathcal{C}_\varepsilon}(r) = O(r^{1/(2\varepsilon)-1})$. Hence $b_\varepsilon = \frac{1}{2\varepsilon} - 1$, the lower bound is tight, and b_ε is continuous at $\varepsilon = \frac{1}{2}$. The class consists of bounded-degree expanders whose vertices are replaced by $\Theta(L) \times \Theta(L)$ grids and whose edges are replaced by matchings of length L between boundary intervals of the grids.

1 Introduction

All graphs are finite and simple. For an n-vertex graph G, a set $Z \subseteq V(G)$ is a *balanced separator* if every component of $G - Z$ has at most $2n/3$ vertices, and $\operatorname{sep}(G)$ denotes the minimum size of a balanced separator. For a class $\mathcal{G}$ of graphs,

$$s_{\mathcal{G}}(n) = \max\{\operatorname{sep}(G) : G \in \mathcal{G},\ |V(G)| \le n\}.$$

An r-*shallow minor* of G is a graph obtained from a subgraph of G by contracting pairwise vertex-disjoint connected subgraphs, each of radius at most r; the *density* of a graph H is $|E(H)|/|V(H)|$. Write $\nabla_r(G)$ for the maximum density of an r-shallow minor of G and

$$\nabla_{\mathcal{G}}(r) = \sup\{\nabla_r(G) : G \in \mathcal{G}\}.$$

A class is *hereditary* if it is closed under induced subgraphs and *monotone* if it is closed under subgraphs. These are exactly the conventions of Dvořák's note [2].

Classes with sublinear separators and classes with polynomial expansion are closely related. Plotkin, Rao and Smith [6] showed that $s_{\mathcal{G}}(n) = \Omega(n^{1-\varepsilon})$ forces $\nabla_{\mathcal{G}}(r) = \Omega(r^{1/(2\varepsilon)-1}/\operatorname{polylog} r)$ [2, Corollary 2], and conversely Dvořák and Norin [3] and Esperet and Raymond [4] showed that $s_{\mathcal{G}}(n) = \Theta(n^{1-\varepsilon})$ for a hereditary class forces $\nabla_{\mathcal{G}}(r) = O(r^{1/\varepsilon-1}\operatorname{polylog} r)$ [2, Theorem 5]. To quantify the gap between the two exponents, Dvořák introduced, for $0 < \varepsilon \le 1$,

$$b_\varepsilon = \sup\{\, b \in \mathbb{R} : \text{ every hereditary class } \mathcal{G} \text{ with } s_{\mathcal{G}}(n) = \Theta(n^{1-\varepsilon}) \text{ satisfies } \nabla_{\mathcal{G}}(r) = \Omega(r^b)\,\}, \quad (1)$$

and proved [2, Corollary 6, Lemmas 10 and 12]

$$\frac{1}{2\varepsilon} - 1 \;\le\; b_\varepsilon \;\le\; \frac{1}{2\varepsilon} - \frac{1}{2} \quad (0 < \varepsilon < \tfrac{1}{2}), \qquad b_\varepsilon = 0 \quad (\tfrac{1}{2} \le \varepsilon \le 1). \quad (2)$$

The upper bound comes from $\lceil m^{\varepsilon/(1-\varepsilon)}\rceil$-fold subdivisions of 3-regular expanders. In his concluding remarks Dvořák writes that it is unclear whether the upper or the lower bound can be improved, observes that $b_{1/2} = 0$ matches the lower bound, and raises the possibility that this is a "dimension 2 artifact" with b_ε discontinuous at $\varepsilon = \frac{1}{2}$. We show that the lower bound is the truth.

Theorem 1.1 (Main theorem)**.** *For every $0 < \varepsilon < \frac{1}{2}$ there is a monotone class $\mathcal{C}_\varepsilon$ of graphs with*

$$s_{\mathcal{C}_\varepsilon}(n) = \Theta(n^{1-\varepsilon}) \qquad \text{and} \qquad \nabla_{\mathcal{C}_\varepsilon}(r) = O(r^{1/(2\varepsilon)-1}).$$

Corollary 1.2. *For $0 < \varepsilon < \frac{1}{2}$, $b_\varepsilon = \frac{1}{2\varepsilon} - 1$. Consequently $\varepsilon \mapsto b_\varepsilon$ is continuous on $(0, 1]$, in particular at $\varepsilon = \frac{1}{2}$. The same value is attained by Dvořák's variant b'_ε, defined as in (1) with $\Theta(n^{1-\varepsilon})$ replaced by $\Omega(n^{1-\varepsilon})$.*

Idea of proof. Corollary 1.2 is a two-line deduction from Theorem 1.1 and Dvořák's Corollary 2; it is carried out in Section 5. For Theorem 1.1, the classes $\mathcal{C}_\varepsilon$ are built from bounded-degree expanders H in which every vertex is replaced by a $(40L+1) \times (40L+1)$ grid (a *block*) and every edge by a perfect matching between two boundary intervals of length L (*ports*) of the corresponding blocks. In Dvořák's subdivided expanders, insulating two branch points by distance L costs L vertices; here it costs L^2 vertices, and this is exactly where the extra $\frac{1}{2}$ in the exponent comes from. Three properties are proved. (i) The whole graph inherits the isoperimetry of H at scale $1/L$, so with g blocks its separators have size $\Theta(gL)$. (ii) Every subgraph on n vertices has a separator of size $O(n/L + \sqrt{n})$: deleting, in every block and for each port, the cheapest of L disjoint crosscuts that cut the port off leaves only planar components, to which the planar separator theorem applies. (iii) Shallow minors of radius $r < 14L$ have bounded density, because their branch sets live in at most two adjacent blocks and a bounded number of blocks embed in a surface of bounded genus, while all minors have density $O(\sqrt{g})$ because the whole graph has Euler genus $O(g)$. Choosing $g = \Theta(L^{1/\varepsilon-2})$ balances (i)–(iii).

2 The construction

Blocks and ports. Fix an integer $L \ge 1$ and let Q_L be the square grid $P_{40L+1} \,\square\, P_{40L+1}$, with vertices (i, j), $0 \le i, j \le 40L$; it has $M_L := (40L+1)^2$ vertices. Its four *ports* $P_1, \dots, P_4$ are the sets of L consecutive boundary vertices centred on the four sides, e.g. $P_1 = \{(0, j) : 20L - \lfloor L/2 \rfloor \le j < 20L - \lfloor L/2 \rfloor + L\}$ on the bottom side. Distinct ports are at grid distance at least $39L \ge 30L$ from each other.

Expanders. Bollobás [1] proved that for every fixed $d \ge 3$ there is a constant $c_d > 0$ such that a uniformly random d-regular graph on g vertices asymptotically almost surely (as $g \to \infty$) has isoperimetric number $\min_{0<|U|\le g/2} |\delta(U)|/|U|$ at least c_d. Since 4-regular graphs on g vertices exist for every $g \ge 5$, there are g_0 and an absolute constant $\alpha > 0$ (any $\alpha < c_4$) such that for every $g \ge g_0$ some 4-regular graph H_g on g vertices satisfies

$$|\delta_{H_g}(U)| \;\ge\; \alpha \min\{|U|,\, g - |U|\} \qquad (U \subseteq V(H_g)), \quad (3)$$

where $\delta_H(U)$ is the set of edges of H with exactly one end in U. Fix such a family once and for all.

The graph $X(g, L)$**.** For $g \geq g_0$ and $L \geq 1$, take a disjoint copy Q_v of Q_L for every $v \in V(H_g)$ and assign its four ports bijectively to the four edges at v. For every edge $uv \in E(H_g)$ join the port of Q_u assigned to uv to the port of Q_v assigned to uv by an *order-preserving* perfect matching (the i-th vertex of one port, in the boundary order, to the i-th vertex of the other). The resulting graph is $X(g, L)$. It has

$$N := |V(X(g, L))| = g\, M_L = \Theta(gL^2) \tag{4}$$

vertices and maximum degree at most 5 (grid degree at most 4, plus at most one matching edge). For $S \subseteq V(X)$ we write $\partial_X S$ for the set of edges of X with exactly one end in S, and $\partial_Q A$ for the analogous set inside a block.

Lemma 2.1. *$X(g, L)$ embeds in a surface of Euler genus at most $g + 1$. Consequently every minor of $X(g, L)$ has density $O(\sqrt{g})$.*

Idea of proof. Draw each block in a disk and each matching inside a band joining the two boundary intervals; the resulting surface is the thickening of H_g, which has g disks and $2g$ bands, so its Euler genus is at most $g + 1$. A simple graph embedded in a surface of Euler genus γ has density $O(\sqrt{\gamma + 1})$, by Euler's formula for many vertices and by $|E| \leq \binom{|V|}{2}$ for few vertices; minors inherit the surface. *[full proof]*

3 Three properties of $X(g, L)$

Throughout, $c_1, c_2, \ldots$ denote absolute positive constants. For $A \subseteq V(Q_L)$ let $\mu(A) = \min\{|A|, M_L - |A|\}$, and say that A has *state* 1 if $|A| \geq M_L/2$ and state 0 otherwise. For a port P_i let $D_i(A)$ be the number of vertices of P_i whose membership in A disagrees with the state of A.

Lemma 3.1 (The block)**.** *There are absolute constants $c_1, C_1 > 0$ such that for every $L \geq 1$ and every $A \subseteq V(Q_L)$:*

- *(a) $|\partial_Q A| \;\geq\; c_1\, \mu(A)/L$;*
- *(b) $D_i(A) \;\leq\; |\partial_Q A| + \mu(A)/L \;\leq\; C_1 |\partial_Q A|$ for each port P_i;*
- *(c) for each port P_i there are L pairwise vertex-disjoint paths $C_i(1), \ldots, C_i(L)$ in Q_L (*crosscuts*), each separating P_i from the other three ports; writing R_i (the *collar *of P_i) for the union of $C_i(L)$ and the set of vertices it cuts off from the other ports, the four collars $R_1, \ldots, R_4$ are pairwise disjoint.*

Idea of proof. (a) is the edge-isoperimetric inequality of the square grid, with $c_1 = 1/41$: if no row and no column is entirely inside the smaller side, every row and every column meeting it contributes a boundary edge, and a set of size μ meets at least $\sqrt{\mu} \geq \mu/(40L + 1)$ rows or columns; if a full row and a full column are inside one side, the same argument applies to the other side. (b): from each disagreeing port vertex draw the perpendicular grid path of L vertices into the grid; these paths are disjoint, and each either contains a boundary edge or consists of L vertices on the minority side. (c): the k-th crosscut is the U-shaped path at distance k around the port; the collars have depth L and width $3L$ and lie far from the corners and from each other. *[full proof]*

Lemma 3.2 (Isoperimetry of X)**.** *There is an absolute $c_2 > 0$ such that for all $g \geq g_0$, $L \geq 1$ and $S \subseteq V(X(g, L))$,*

$$|\partial_X S| \;\geq\; \frac{c_2}{L} \min\{|S|,\, N - |S|\}.$$

Idea of proof. Let U be the set of blocks in which S has state 1. Across each edge of $\delta_{H_g}(U)$ run L matching edges; each of them is either in $\partial_X S$ or has a disagreeing endpoint in one of its two blocks, and Lemma 3.1(b) bounds the number of disagreeing port vertices by C_1 times the boundary inside the block. Summing, $L|\delta_{H_g}(U)| \leq (1 + 4C_1)|\partial_X S|$. Also $\sum_v \mu(S \cap V(Q_v)) \leq (L/c_1)|\partial_X S|$ by Lemma 3.1(a). Since $|S|$ differs from $M_L|U|$ by at most $\sum_v \mu(S \cap V(Q_v))$, the expander inequality (3) for U and $M_L = \Theta(L^2)$ give the claim. *[full proof]*

Corollary 3.3. $\operatorname{sep}(X(g,L)) \ \geq \ c_3\, N/L = \Omega(gL)$.

Idea of proof. If a balanced separator Z has fewer than $N/6$ vertices, some union A of components of $X - Z$ has $N/6 \leq |A| \leq N/2$; all edges leaving A end in Z, so $5|Z| \geq |\partial_X A| \geq c_2 N/(6L)$. *[full proof]*

Lemma 3.4 (Hereditary separators)**.** *Every subgraph F of $X(g,L)$ with n vertices has a balanced separator of size at most $4n/L + c_4\sqrt{n}$.*

Idea of proof. In each block and for each port choose, among the L disjoint crosscuts of Lemma 3.1(c), one containing the fewest vertices of F; the chosen crosscuts contain at most $4n/L$ vertices of F in total. After deleting them every component of F lies either in the part of a block outside its collars, or in two collars joined by one matching; both are planar. If some component still has more than $2n/3$ vertices, add a planar separator of it. *[full proof]*

Lemma 3.5 (Shallow minors below the block scale)**.** *There is an absolute C_5 such that for all $g \geq g_0$, $L \geq 1$ and $0 \leq r < 14L$, every r-shallow minor J of $X(g,L)$ satisfies $|E(J)| \leq C_5 |V(J)|$.*

Idea of proof. Any path between two distinct ports of one block, or between two non-adjacent blocks, has length at least $30L$. Hence a branch set of radius $r < 14L$ lies in at most two adjacent blocks, and adjacent branch sets have centres in the same or in adjacent blocks. Orient J arbitrarily and charge each edge to the block v containing the centre of its tail. The edges charged to v, together with their ends, form a minor of the subgraph of X induced by the at most 17 blocks within H_g-distance 2 of v, which embeds in a surface of bounded genus and therefore has bounded density by Lemma 2.1. Each vertex of J is an end of edges charged to at most 5 blocks. *[full proof]*

Corollary 3.6. *For all $g \geq g_0$, $L \geq 1$ and $r \geq 0$,*

$$\nabla_r(X(g,L)) \leq \begin{cases} C_5 & \text{if } r < 14L, \\ C_6\sqrt{g} & \text{for all } r, \end{cases}$$

with absolute constants C_5, C_6.

Proof. The first line is Lemma 3.5; the second is Lemma 2.1, since an r-shallow minor is a minor. □

4 Proof of the main theorem

Fix $0 < \varepsilon < \frac{1}{2}$ and put

$$p = \frac{1}{\varepsilon} - 2 > 0. \tag{5}$$

For every $L \geq L_0$, where L_0 is large enough that $\lceil L_0^p \rceil \geq g_0$, let $g_L = \lceil L^p \rceil$ and $X_L = X(g_L, L)$. Let $\mathcal{C}_\varepsilon$ be the class of all subgraphs of all the graphs X_L, $L \geq L_0$; it is monotone by definition. By (4) and (5),

$$N_L := |V(X_L)| = \lceil L^p \rceil (40L+1)^2 = \Theta(L^{p+2}) = \Theta(L^{1/\varepsilon}), \tag{6}$$

N_L is strictly increasing in L, and $N_{L+1}/N_L \to 1$.

Separators: lower bound. By Corollary 3.3, (6) and $(p+1)/(p+2) = 1 - \varepsilon$,

$$\operatorname{sep}(X_L) = \Omega(g_L L) = \Omega(L^{p+1}) = \Omega(N_L^{(p+1)/(p+2)}) = \Omega(N_L^{1-\varepsilon}). \tag{7}$$

Given $n \geq N_{L_0}$ let L be maximal with $N_L \leq n$. Then $X_L \in \mathcal{C}_\varepsilon$ has at most n vertices and $n < N_{L+1} = (1+o(1))N_L$, so $s_{\mathcal{C}_\varepsilon}(n) \geq \operatorname{sep}(X_L) = \Omega(n^{1-\varepsilon})$.

Separators: upper bound. Let $F \in \mathcal{C}_\varepsilon$ have n vertices, say $F \subseteq X_L$. Then $n \le N_L = O(L^{1/\varepsilon})$, i.e. $n^\varepsilon = O(L)$, so $n/L = O(n^{1-\varepsilon})$; and $\sqrt{n} \le n^{1-\varepsilon}$ because $\varepsilon \le \frac{1}{2}$. Lemma 3.4 gives $\mathrm{sep}(F) \le 4n/L + c_4\sqrt{n} = O(n^{1-\varepsilon})$, with a constant independent of L. Hence $s_{\mathcal{C}_\varepsilon}(n) = O(n^{1-\varepsilon})$, and altogether

$$s_{\mathcal{C}_\varepsilon}(n) = \Theta(n^{1-\varepsilon}). \tag{8}$$

Expansion. An r-shallow minor of a subgraph of X_L is an r-shallow minor of X_L. By Corollary 3.6, if $L > r/14$ its density is at most C_5, and if $L \le r/14$ its density is at most $C_6\sqrt{g_L} = C_6\sqrt{\lceil L^p \rceil} \le C_7\,(r/14)^{p/2}$. Since $p/2 = \frac{1}{2\varepsilon} - 1$,

$$\nabla_{\mathcal{C}_\varepsilon}(r) = O(r^{p/2}) = O(r^{1/(2\varepsilon)-1}). \tag{9}$$

Equations (8) and (9) prove Theorem 1.1. □

5 The value of b_ε

Proof of Corollary 1.2. Let B be the set of real numbers b such that every hereditary class $\mathcal{G}$ with $s_{\mathcal{G}}(n) = \Theta(n^{1-\varepsilon})$ satisfies $\nabla_{\mathcal{G}}(r) = \Omega(r^b)$, so that $b_\varepsilon = \sup B$.

Upper bound. Let $b > \frac{1}{2\varepsilon} - 1$. The class $\mathcal{C}_\varepsilon$ of Theorem 1.1 is monotone, hence hereditary, and satisfies $s_{\mathcal{C}_\varepsilon}(n) = \Theta(n^{1-\varepsilon})$. Since $\nabla_{\mathcal{C}_\varepsilon}(r) = O(r^{1/(2\varepsilon)-1})$ and $r^{1/(2\varepsilon)-1}/r^b \to 0$, the relation $\nabla_{\mathcal{C}_\varepsilon}(r) = \Omega(r^b)$ fails, so $b \notin B$. Thus every element of B is at most $\frac{1}{2\varepsilon} - 1$ and $b_\varepsilon \le \frac{1}{2\varepsilon} - 1$.

Lower bound. Let $b < \frac{1}{2\varepsilon} - 1$ and let $\mathcal{G}$ be a hereditary class with $s_{\mathcal{G}}(n) = \Theta(n^{1-\varepsilon})$. In particular $s_{\mathcal{G}}(n) = \Omega(n^{1-\varepsilon})$, and Dvořák's Corollary 2 [2] gives $\nabla_{\mathcal{G}}(r) = \Omega(r^{1/(2\varepsilon)-1}/\operatorname{polylog} r) = \Omega(r^b)$, the polylogarithmic loss being absorbed by the strict inequality. Hence $b \in B$ and $b_\varepsilon \ge \frac{1}{2\varepsilon} - 1$.

Therefore $b_\varepsilon = \frac{1}{2\varepsilon} - 1$ on $(0, \frac{1}{2})$. As $\frac{1}{2\varepsilon} - 1 \to 0$ when $\varepsilon \to \frac{1}{2}^-$ and $b_\varepsilon = 0$ on $[\frac{1}{2}, 1]$ by (2), b_ε is continuous at $\frac{1}{2}$ (and trivially elsewhere).

For the variant b'_ε, Dvořák observes $\max\{\frac{1}{2\varepsilon} - 1, 0\} \le b'_\varepsilon \le b_\varepsilon$ [2, concluding remarks]; combined with $b_\varepsilon = \frac{1}{2\varepsilon} - 1$ this gives $b'_\varepsilon = \frac{1}{2\varepsilon} - 1$. □

6 Remarks

Remark 6.1 (Comparison with Dvořák's construction). Dvořák's upper bound $b_\varepsilon \le \frac{1}{2\varepsilon} - \frac{1}{2}$ uses subdivided cubic expanders: a vertex set of size m with separators $\Theta(m^{1-\varepsilon})$ is obtained by subdividing each edge $\Theta(m^{\varepsilon/(1-\varepsilon)})$ times, and a shallow minor of radius r can already contract whole subdivided edges once r exceeds the subdivision length. In $X(g, L)$ the insulating distance L between ports costs L^2 vertices per block instead of L per edge, so the radius at which minors start to see the expander is $\Theta(L) = \Theta(N^\varepsilon)$ rather than $\Theta(N^{\varepsilon/(1-\varepsilon)})$, while the separator size stays $\Theta(N^{1-\varepsilon})$. This is consistent with Dvořák's remark that $b_{1/2} = 0$ (planar-like behaviour) might be a "dimension 2" phenomenon: two-dimensional blocks are exactly what is needed to make the phenomenon persist for all $\varepsilon < \frac{1}{2}$.

Proposition 6.2 (Exact-order convention)**.** *Define* $\tilde{s}_{\mathcal{G}}(n) = \max\{\mathrm{sep}(G) : G \in \mathcal{G},\ |V(G)| = n\}$. *Let* $\mathcal{C}'_\varepsilon = \{F \sqcup I_k : F \in \mathcal{C}_\varepsilon,\ k \ge 0\}$, *where* I_k *is the edgeless graph on* k *vertices. Then* $\mathcal{C}'_\varepsilon$ *is monotone,* $\nabla_{\mathcal{C}'_\varepsilon} = \nabla_{\mathcal{C}_\varepsilon}$, *and* $\tilde{s}_{\mathcal{C}'_\varepsilon}(n) = \Theta(n^{1-\varepsilon})$.

Idea of proof. Dvořák's $s_{\mathcal{G}}$ is defined with $|V(G)| \le n$, so this proposition is not needed for Corollary 1.2; it shows the result is insensitive to the convention. Isolated vertices do not change shallow-minor densities or separators, and for a given n the graph $X_L \sqcup I_{n-N_L}$ with $N_L \le n < N_{L+1}$ has $n = (1 + o(1))N_L$ vertices; the argument of Corollary 3.3 tolerates the slightly weaker balance condition $\frac{2}{3}n \le 0.7N_L$. *[full proof]*

Remark 6.3 (Constants). All constants are absolute except where they visibly depend on ε through p. The thresholds "$L \geq L_0$" and "$r < 14L$" are explicit; the referee's computations on the instance $g = 6$, $L = 3$ (Appendix B) found the locality claims of Lemma 3.5 to hold with a large margin (balls of radius up to $19L$ meet at most two blocks).

Remark 6.4 (Novelty). The referee's search (Appendix B) found no work resolving the question; the only citing papers concern other topics. This has not been checked by a human.

A Full proofs

Proof of Lemma 2.1. For each $v \in V(H_g)$ draw Q_v in a closed disk Δ_v so that the boundary cycle of the grid lies on the boundary circle. For each edge uv attach a band (a copy of $[0,1] \times [0,1]$) to $\partial\Delta_u$ along the boundary interval spanned by the port of Q_u assigned to uv and to $\partial\Delta_v$ along the corresponding interval; draw the L matching edges as disjoint straight segments across the band, which is possible because the matching is order-preserving. The union Σ_0 of the g disks and $2g$ bands (recall $|E(H_g)| = 4g/2 = 2g$) is a compact surface with boundary, and $X(g, L)$ is drawn in it without crossings. Its Euler characteristic is $\chi(\Sigma_0) = g - 2g = -g$ (each band attached along two intervals lowers χ by one). Capping each of the $\beta \geq 1$ boundary circles with a disk yields a closed surface Σ with $\chi(\Sigma) = -g + \beta \geq 1 - g$, hence Euler genus $\mathrm{eg}(\Sigma) = 2 - \chi(\Sigma) \leq g + 1$.

Now let J be a simple graph embedded in a surface of Euler genus γ, with $q = |V(J)| \geq 3$ vertices. Euler's formula gives $|E(J)| \leq 3q - 6 + 3\gamma$. If $q \geq \sqrt{\gamma+1}$ then $|E(J)|/q \leq 3 + 3\gamma/q \leq 3 + 3\sqrt{\gamma+1}$; otherwise $|E(J)|/q \leq (q-1)/2 < \sqrt{\gamma+1}/2$. For $q \leq 2$ the density is at most $\frac{1}{2}$. Hence the density is at most $3 + 3\sqrt{\gamma+1} = O(\sqrt{\gamma+1})$. A minor of a graph embedded in Σ embeds in Σ (delete, and contract edges within the drawing), so every minor of $X(g, L)$ has density at most $3 + 3\sqrt{g+2} = O(\sqrt{g})$. $\square$

Lemma 3.1 (The block). *There are absolute constants $c_1, C_1 > 0$ such that for every $L \geq 1$ and every $A \subseteq V(Q_L)$:*

(a) $|\partial_Q A| \;\geq\; c_1\, \mu(A)/L$;

(b) $D_i(A) \;\leq\; |\partial_Q A| + \mu(A)/L \;\leq\; C_1 |\partial_Q A|$ *for each port P_i;*

(c) for each port P_i there are L pairwise vertex-disjoint paths $C_i(1), \dots, C_i(L)$ in Q_L (crosscuts)*, each separating P_i from the other three ports; writing R_i (the* collar *of P_i) for the union of $C_i(L)$ and the set of vertices it cuts off from the other ports, the four collars $R_1, \dots, R_4$ are pairwise disjoint.*

Proof. Write $m = 40L + 1$ for the side of the grid and $M = m^2$.

(a) We show $|\partial_Q A| \geq \mu(A)/m$, so $c_1 = 1/41$ works. Since $\partial_Q A = \partial_Q(V \setminus A)$ and $\mu(A) = \mu(V \setminus A)$, we may assume $|A| \leq M/2$, so $\mu(A) = |A|$. Call a row or column *A-full* if all its vertices are in A.

Case 1: no row and no column is A-full. Every row that meets A contains a vertex of A and a vertex not in A, hence (rows are paths) two consecutive vertices, one in A and one not: an edge of $\partial_Q A$ lying in that row. Distinct rows give distinct edges, so $|\partial_Q A| \geq \rho$, the number of rows meeting A; likewise $|\partial_Q A| \geq \kappa$, the number of columns meeting A. As A lies in the $\rho \times \kappa$ sub-array of those rows and columns, $|A| \leq \rho\kappa \leq \max\{\rho, \kappa\}^2$, so $|\partial_Q A| \geq \sqrt{|A|} \geq |A|/m$ because $|A| \leq M = m^2$.

Case 2: some row is A-full but no column is. Then every column meets A and is not A-full, so as above every column contributes a boundary edge and $|\partial_Q A| \geq m \geq |A|/m$. The case of an A-full column and no A-full row is symmetric.

Case 3: some row and some column are A-full. Put $B = V \setminus A$; then $|B| \geq M/2 \geq |A|$. No row is B-full (it meets the A-full column) and no column is B-full. Applying Case 1 to B gives $|\partial_Q A| = |\partial_Q B| \geq \sqrt{|B|} \geq |B|/m \geq |A|/m$.

(b) By symmetry let P_i be the bottom port, $P_i = \{(0, j) : a \leq j < a + L\}$. Suppose first that A has state 1, so the disagreeing port vertices are those of $P_i \setminus A$ and $\mu(A) = |V \setminus A|$. For each

$(0,j) \in P_i \setminus A$ consider the vertical path $\pi_j = (0,j),(1,j),\dots,(L-1,j)$ of L vertices (it exists since $L-1 \le 40L$). The paths π_j are pairwise vertex-disjoint. If π_j meets A, then since its first vertex is not in A, some edge of π_j has exactly one end in A, i.e. belongs to $\partial_Q A$; distinct paths give distinct edges. If π_j does not meet A, its L vertices lie in $V \setminus A$, and the paths being disjoint, there are at most $|V \setminus A|/L$ such j. Hence $D_i(A) = |P_i \setminus A| \le |\partial_Q A| + \mu(A)/L$. If A has state 0, apply the same argument to $V \setminus A$, which has state 1 (if $|A| < M/2$ then $|V \setminus A| > M/2$), the same boundary, the same μ, and the same disagreeing vertices. Finally $\mu(A)/L \le |\partial_Q A|/c_1$ by (a), so $C_1 = 1 + 1/c_1 = 42$ works.

(c) Again let $P_i = \{(0,j) : a \le j < a+L\}$ with $a = 20L - \lfloor L/2 \rfloor$, so $a \ge 19L$ and $a+L-1 \le 21L$. For $1 \le k \le L$ let $C_i(k)$ be the path

$$(0,a-k),(1,a-k),\dots,(k,a-k),\ (k,a-k+1),\dots,(k,a+L-1+k),\ (k-1,a+L-1+k),\dots,(0,a+L-1+k),$$

a U-shape using column $a-k$ and column $a+L-1+k$ in rows $0,\dots,k$ and row k in columns $a-k,\dots,a+L-1+k$. All these vertices exist since $a-L \ge 18L \ge 0$ and $a+2L-1 \le 22L \le 40L$. For $k \ne k'$ the paths are disjoint: they use different columns for their vertical parts and different rows for their horizontal parts, and a vertical part of $C_i(k)$ (rows $\le k$, column $a-k$ or $a+L-1+k$) meets the horizontal part of $C_i(k')$ (row k', columns $a-k',\dots,a+L-1+k'$) only if $k' \le k$ and $a-k' \le a-k$, i.e. $k' \ge k$, so $k=k'$; the symmetric case is identical. Let $R_i(k)$ be the set of vertices (x,y) with $x<k$ and $a-k<y<a+L-1+k$. Then $P_i \subseteq R_i(1) \subseteq R_i(k)$, and every grid edge with one end in $R_i(k)$ has its other end in $R_i(k) \cup C_i(k)$: a neighbour of $(x,y) \in R_i(k)$ outside $R_i(k)$ has $x=k$ (then it is on row k within the column range, in $C_i(k)$) or $y \in \{a-k, a+L-1+k\}$ with $x<k$ (then it is on a vertical part of $C_i(k)$); $x<0$ is impossible. Hence every path from P_i to a vertex outside $R_i(k) \cup C_i(k)$ passes through $C_i(k)$. The collar $R_i = R_i(L) \cup C_i(L)$ is contained in rows $0,\dots,L$ and columns $a-L,\dots,a+2L-1 \subseteq [18L,22L]$; the other three ports are on the other three sides, at rows $\ge 20L - \lfloor L/2 \rfloor > L$ or at columns ≤ 0 or $\ge 40L$ outside $[18L,22L]$, so they lie outside R_i and each $C_i(k)$ separates P_i from them. By symmetry the collar of the left port lies in columns $0,\dots,L$ and rows in $[18L,22L]$, and similarly for the other two; since $L < 18L$, the four collars are pairwise disjoint. $\square$

Lemma 3.2 (Isoperimetry of X). *There is an absolute $c_2 > 0$ such that for all $g \ge g_0$, $L \ge 1$ and $S \subseteq V(X(g,L))$,*

$$|\partial_X S| \ \ge\ \frac{c_2}{L}\min\{|S|,\ N-|S|\}.$$

Proof. For $v \in V(H_g)$ let $S_v = S \cap V(Q_v)$, $s_v = |S_v|$, $\mu_v = \mu(S_v)$, and let b_v be the number of edges of $\partial_X S$ that lie inside Q_v; the edges of $\partial_X S$ are either inside some block or matching edges, so

$$\sum_v b_v + \#\{\text{matching edges in } \partial_X S\} = |\partial_X S|. \tag{10}$$

By Lemma 3.1(a) applied in Q_v, $b_v = |\partial_Q S_v| \ge c_1 \mu_v / L$, hence

$$\sum_v \mu_v \le \frac{L}{c_1}\sum_v b_v \le \frac{L}{c_1}|\partial_X S|. \tag{11}$$

Let $U = \{v : s_v \ge M_L/2\}$ be the set of blocks in which S_v has state 1. Consider an edge $uv \in \delta_{H_g}(U)$ with $u \in U$, $v \notin U$, and the L matching edges pp' between the corresponding ports, $p \in Q_u$, $p' \in Q_v$. If $pp' \notin \partial_X S$ then either $p,p' \in S$, in which case $p' \in S_v$ disagrees with the state 0 of S_v, or $p,p' \notin S$, in which case $p \notin S_u$ disagrees with the state 1 of S_u. Hence, with c_{uv} the number of matching edges of this interface in $\partial_X S$ and using Lemma 3.1(b),

$$L \le c_{uv} + D(S_u) + D(S_v) \le c_{uv} + C_1(b_u + b_v),$$

where $D(\cdot)$ refers to the relevant port. Summing over $uv \in \delta_{H_g}(U)$, each block has four ports so each b_v appears at most four times, and by (10)

$$L\,|\delta_{H_g}(U)| \le \sum_{uv\in\delta(U)} c_{uv} + 4C_1 \sum_v b_v \le (1+4C_1)|\partial_X S|. \tag{12}$$

Now $|S| = \sum_{v\in U}(M_L - \mu_v) + \sum_{v\notin U}\mu_v$, so $\big||S| - M_L|U|\big| \le \sum_v \mu_v$, and likewise $\big|(N-|S|) - M_L(g-|U|)\big| \le \sum_v \mu_v$. Therefore, using (3), (12), (11) and $M_L = (40L+1)^2 \le 1681L^2$,

$$\begin{aligned}\min\{|S|, N-|S|\} &\le M_L \min\{|U|, g-|U|\} + \sum_v \mu_v \le \frac{M_L}{\alpha}|\delta_{H_g}(U)| + \sum_v \mu_v \\ &\le \Big(\frac{1681(1+4C_1)}{\alpha} + \frac{1}{c_1}\Big) L\,|\partial_X S|.\end{aligned}$$

This is the claim with $c_2 = \big(1681(1+4C_1)/\alpha + 1/c_1\big)^{-1}$. □

Corollary 3.3. $\operatorname{sep}(X(g,L)) \ \ge\ c_3\, N/L = \Omega(gL)$.

Proof. Let Z be a balanced separator of $X = X(g,L)$. If $|Z| \ge N/6$ we are done (with $c_3 \le 1/6$, as $L \ge 1$). Otherwise the components of $X - Z$ have at most $2N/3$ vertices each and more than $5N/6$ vertices in total. If some component K has $|K| \ge N/2$, let A be the union of the other components: $|A| > 5N/6 - 2N/3 = N/6$ and $|A| \le N - |K| \le N/2$. If some component K has $N/6 \le |K| < N/2$, let $A = K$. Otherwise all components have fewer than $N/6$ vertices; add them one at a time until the total first reaches $N/6$, obtaining A with $N/6 \le |A| < N/3$. In all cases A is a union of components of $X - Z$ with $N/6 \le |A| \le N/2$, so $\min\{|A|, N-|A|\} \ge N/6$, and every edge leaving A ends in Z. Since X has maximum degree at most 5, Lemma 3.2 gives $5|Z| \ge |\partial_X A| \ge c_2 N/(6L)$, i.e. $|Z| \ge c_2 N/(30L)$. Hence $c_3 = \min\{1/6, c_2/30\}$ works, and $N/L = gM_L/L = \Omega(gL)$. □

Lemma 3.4 (Hereditary separators)**.** *Every subgraph F of $X(g,L)$ with n vertices has a balanced separator of size at most $4n/L + c_4\sqrt{n}$.*

Proof. Let $F \subseteq X(g,L)$ with $n = |V(F)|$ and $n_v = |V(F) \cap V(Q_v)|$. For each block Q_v and each of its ports P_i, the L crosscuts $C_i(1), \dots, C_i(L)$ of Lemma 3.1(c) are pairwise disjoint subsets of $V(Q_v)$, so one of them, $C_{v,i}$, satisfies $|V(F) \cap C_{v,i}| \le n_v/L$. Let $Z = \bigcup_{v,i}(V(F) \cap C_{v,i})$; then $|Z| \le \sum_v 4n_v/L = 4n/L$.

Consider a component K of $F - Z$. Inside a block Q_v, the vertices of K avoid the four chosen crosscuts, so each vertex of $K \cap V(Q_v)$ lies either in the inner region $R_{v,i}$ enclosed by P_i and $C_{v,i}$ for some i (call it the collar interior of port i), or in the *core* $V(Q_v) \setminus \bigcup_i (R_{v,i} \cup C_{v,i})$. By Lemma 3.1(c), no edge of Q_v joins a collar interior to the core or to another collar interior, and the only edges of X leaving Q_v are matching edges at the ports, which lie in the collar interiors. Consequently, if K contains a core vertex of Q_v then K is contained in the core of Q_v, a subgraph of a planar grid. Otherwise K is contained in collar interiors; a collar interior of Q_v at port i is joined by matching edges only to the collar interior of the neighbouring block at the matched port, and the matched collar interior is separated from the rest of its block by its own crosscut. Hence K is contained in the union of two matched collar interiors and the matching between them. This graph is planar: draw the two collars in two disjoint disks with the ports on the boundaries and the matching edges as disjoint segments in a band joining them, as in the proof of Lemma 2.1; a disk with one band attached is a disk, i.e. the sphere with a hole. So every component of $F - Z$ is planar.

If every component of $F - Z$ has at most $2n/3$ vertices, Z is a balanced separator of F of size at most $4n/L$. Otherwise exactly one component K has more than $2n/3$ vertices (two such components would exceed n). By the planar separator theorem of Lipton and Tarjan [5], K has a set Y of at most $c_4\sqrt{|K|} \le c_4\sqrt{n}$ vertices such that every component of $K - Y$ has at most $\frac{2}{3}|K| \le \frac{2}{3}n$ vertices. Every component of $F - (Z \cup Y)$ is either a component of $F - Z$ other than K, of size at most $n - |K| < n/3$, or a component of $K - Y$. Thus $Z \cup Y$ is a balanced separator of F of size at most $4n/L + c_4\sqrt{n}$. □

Lemma 3.5 (Shallow minors below the block scale)**.** *There is an absolute C_5 such that for all $g \geq g_0$, $L \geq 1$ and $0 \leq r < 14L$, every r-shallow minor J of $X(g, L)$ satisfies $|E(J)| \leq C_5|V(J)|$.*

Proof. Write $X = X(g, L)$ and $H = H_g$. For a vertex $z \in V(X)$ let $\beta(z) \in V(H)$ denote the block containing it. We first establish two distance facts.

Claim A.1. *(i) If $P \neq P'$ are distinct ports of the same block Q_u, then $\operatorname{dist}_X(P, P') \geq 30L$. (ii) If $u, w \in V(H)$ are distinct and non-adjacent, then $\operatorname{dist}_X(V(Q_u), V(Q_w)) \geq 30L$.*

Proof of the claim. Let π be a shortest path between the two sets in question, chosen among shortest paths to minimise the number of matching edges used, and let $u = w_0, w_1, \ldots, w_t$ be the sequence of blocks it visits (consecutive terms distinct, each change of block along a matching edge). We claim $w_{i+1} \neq w_{i-1}$ for all i. Indeed, if $w_{i+1} = w_{i-1}$, then π enters Q_{w_i} from $Q_{w_{i-1}}$ along a matching edge qq' and later returns along a matching edge $\tilde{q}'\tilde{q}$ with $\tilde{q} \in Q_{w_{i-1}}$; because H is simple there is only one edge $w_{i-1}w_i$, so $q, \tilde{q}$ lie in the same port P of $Q_{w_{i-1}}$ and $q', \tilde{q}'$ in the matched port P'. The excursion from q to $\tilde{q}$ has length at least $2 + \operatorname{dist}_{Q_{w_i}}(q', \tilde{q}') = 2 + \operatorname{dist}_{Q_{w_{i-1}}}(q, \tilde{q})$, because the matching is order-preserving and both ports are straight segments of the grid boundary, along which grid distance equals the difference of positions. Replacing the excursion by the path along P from q to $\tilde{q}$ gives a strictly shorter walk between the same endpoints, contradicting minimality. Hence there is no backtracking.

(i) If $t = 0$, π stays in Q_u and has length at least $\operatorname{dist}_{Q_u}(P, P') \geq 39L \geq 30L$. If $t \geq 1$, then π ends in $Q_u = Q_{w_t}$, so $t \geq 2$ (as $w_1 \neq w_0$), and w_1 is an intermediate block entered from w_0 and left towards $w_2 \neq w_0$; the entering and leaving matching edges belong to different edges of H at w_1, hence to distinct ports of Q_{w_1}, and the segment of π inside Q_{w_1} between them has length at least $39L \geq 30L$.

(ii) Here $w_t = w$ with $w \neq u$ non-adjacent, so $t \geq 2$, and the same argument applied to the intermediate block w_1 gives length at least $30L$. □

Now let J be an r-shallow minor of X with $r < 14L$, with branch sets B_x $(x \in V(J))$, pairwise disjoint connected subgraphs of X of radius at most r; choose centres $c_x \in V(B_x)$ with every vertex of B_x at distance at most r from c_x (in B_x, hence in X), and put $\beta(x) = \beta(c_x)$.

Claim A.2. *(a) $V(B_x) \subseteq V(Q_{\beta(x)}) \cup V(Q_w)$ for some $w \in N_H(\beta(x))$. (b) If $xy \in E(J)$ then $\beta(y) \in N_H[\beta(x)]$. (c) If $xy \in E(J)$ and $\beta(x) = v$, then $V(B_x) \cup V(B_y) \subseteq \bigcup\{V(Q_w) : w \in N_H^2[v]\}$, where $N_H^2[v]$ is the set of vertices at H-distance at most 2 from v.*

Proof of the claim. (a) Let $z \in V(B_x)$ with $\beta(z) \neq \beta(x)$, and let π be a shortest c_x–z path in X, of length at most $r < 14L < 30L$, chosen to minimise the number of matching edges. By Claim A.1(ii), $\beta(z) \in N_H(\beta(x))$. By the no-backtracking property established in the proof of Claim A.1, the block sequence of π is $\beta(x), w_1, \ldots, w_t = \beta(z)$ without backtracking, and $t \geq 2$ would produce a segment of length at least $30L$ inside Q_{w_1}; so $t = 1$ and π leaves $Q_{\beta(x)}$ exactly once, through the port P_{w_1} assigned to the edge $\beta(x)w_1$. If $z' \in V(B_x)$ is another vertex with $\beta(z') = w' \notin \{\beta(x), w_1\}$, the corresponding path leaves $Q_{\beta(x)}$ through the port $P_{w'} \neq P_{w_1}$, and the concatenation of the two paths through c_x shows $\operatorname{dist}_X(P_{w_1}, P_{w'}) \leq 2r < 28L < 30L$, contradicting Claim A.1(i).

(b) An edge of J between x and y comes from an edge of X between B_x and B_y, so $\operatorname{dist}_X(c_x, c_y) \leq 2r + 1 < 28L + 1 \leq 30L$ (as $L \geq 1$). By Claim A.1(ii), $\beta(y) \in N_H[\beta(x)]$.

(c) By (a), B_x lies in Q_v and possibly one block adjacent to v; by (b) $\beta(y) \in N_H[v]$, and by (a) B_y lies in $Q_{\beta(y)}$ and possibly one block adjacent to $\beta(y)$, all of which are within H-distance 2 of v. □

Orient the edges of J arbitrarily. For $v \in V(H)$ let E_v be the set of oriented edges of J whose tail x has $\beta(x) = v$, and let W_v be the set of ends (tails and heads) of the edges in E_v. Then $\sum_v |E_v| = |E(J)|$. Let X_v be the subgraph of X induced by $\bigcup\{V(Q_w) : w \in N_H^2[v]\}$; as H is 4-regular, $|N_H^2[v]| \leq 1 + 4 + 12 = 17$. By Claim A.2(c), for every $x \in W_v$ the branch set B_x is a connected subgraph of X_v (all its vertices, hence all its edges, lie in X_v), and every edge of E_v is realised by an edge of X_v between two such branch sets. Hence the graph (W_v, E_v) is a minor of X_v. The graph X_v embeds in the surface obtained from at most 17 disks and the bands corresponding to edges of H within $N_H^2[v]$, at most $17 \cdot 4/2 = 34$ bands, exactly as in the proof of Lemma 2.1; its Euler genus is therefore at most 35, and by the density bound in that proof $|E_v| \le (3 + 3\sqrt{36})|W_v| = 21|W_v|$.

Finally, a vertex $x \in V(J)$ belongs to W_v only if x is the tail of an edge in E_v, in which case $v = \beta(x)$, or the head of an edge $yx \in E_v$, in which case $v = \beta(y) \in N_H[\beta(x)]$ by Claim A.2(b). Thus x lies in W_v for at most $|N_H[\beta(x)]| = 5$ blocks v, and

$$|E(J)| = \sum_v |E_v| \le 21 \sum_v |W_v| \le 105\,|V(J)|.$$

So $C_5 = 105$ works. □

Proposition 6.2 (Exact-order convention). *Define $\tilde{s}_{\mathcal{G}}(n) = \max\{\mathrm{sep}(G) : G \in \mathcal{G},\ |V(G)| = n\}$. Let $\mathcal{C}'_\varepsilon = \{F \sqcup I_k : F \in \mathcal{C}_\varepsilon,\ k \ge 0\}$, where I_k is the edgeless graph on k vertices. Then $\mathcal{C}'_\varepsilon$ is monotone, $\nabla_{\mathcal{C}'_\varepsilon} = \nabla_{\mathcal{C}_\varepsilon}$, and $\tilde{s}_{\mathcal{C}'_\varepsilon}(n) = \Theta(n^{1-\varepsilon})$.*

Proof. Monotone. A subgraph of $F \sqcup I_k$ is $F' \sqcup I_{k'}$ where F' is a subgraph of F (a vertex of F that becomes isolated may be counted in the second part), and $F' \in \mathcal{C}_\varepsilon$ since $\mathcal{C}_\varepsilon$ is monotone.

Expansion. Branch sets of a minor model are connected, so an isolated vertex can only be a branch set on its own, incident with no edge of the minor. Hence an r-shallow minor of $F \sqcup I_k$ is $J \sqcup I_{k'}$ with J an r-shallow minor of F (possibly empty), and its density is at most that of J. Therefore $\nabla_{\mathcal{C}'_\varepsilon}(r) = \nabla_{\mathcal{C}_\varepsilon}(r)$ for every r.

Upper bound. Let $G = F \sqcup I_k$ have $n \ge 2$ vertices and let Z be a minimum balanced separator of F. Every component of $G - Z$ is a component of $F - Z$, of order at most $\frac{2}{3}|V(F)| \le \frac{2}{3}n$, or a single isolated vertex. So Z is a balanced separator of G and $\mathrm{sep}(G) \le \mathrm{sep}(F) = O(|V(F)|^{1-\varepsilon}) = O(n^{1-\varepsilon})$ by (8).

Lower bound. Let $n \ge N_{L_0}$, let L be maximal with $N_L \le n$, put $k = n - N_L < N_{L+1} - N_L$ and $G_n = X_L \sqcup I_k \in \mathcal{C}'_\varepsilon$. Since $N_{L+1}/N_L \to 1$ we have $n = (1 + o(1))N_L$, so for all n large enough $\frac{2}{3}n \le 0.7N_L$. Let Z be a balanced separator of G_n and $Z' = Z \cap V(X_L)$. Every component of $X_L - Z'$ is a component of $G_n - Z$, hence has at most $0.7N_L$ vertices. If $|Z'| \ge N_L/6$ we are done. Otherwise $X_L - Z'$ has more than $\frac{5}{6}N_L$ vertices, and as in the proof of Corollary 3.3 there is a union A of components of $X_L - Z'$ with $N_L/6 \le |A| \le 0.7N_L$: if some component has at least $N_L/6$ vertices take it alone (it has at most $0.7N_L$); otherwise add components of order $< N_L/6$ one at a time until the total first reaches $N_L/6$, when it is below $N_L/3$. Then $\min\{|A|, N_L - |A|\} \ge \min\{N_L/6, 0.3N_L\} = N_L/6$, every edge of X_L leaving A ends in Z', and Lemma 3.2 gives $5|Z'| \ge |\partial_X A| \ge c_2 N_L/(6L)$. Hence

$$|Z| \ge |Z'| \ge \frac{c_2 N_L}{30L} = \Omega(g_L L) = \Omega(N_L^{1-\varepsilon}) = \Omega(n^{1-\varepsilon})$$

by (7) and $n = (1 + o(1))N_L$. Together with the upper bound, $\tilde{s}_{\mathcal{C}'_\varepsilon}(n) = \Theta(n^{1-\varepsilon})$; the finitely many $n < N_{L_0}$ are irrelevant to a Θ statement. Thus Theorem 1.1, and with it Corollary 1.2, hold for $\mathcal{C}'_\varepsilon$ under the exact-order convention as well. □

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (`claude-fable-5`, 2026-09-02) on the *original* writeup. Its step numbers (S1–S20) refer to that writeup, not to this note; the two items it labels GAP are the ones repaired in Section 5 and Proposition 6.2. The scripts it mentions are in `verification/scripts/2001.09679__00/`. Treat it as machine output, not as an authority.

Reviewed files:

- Writeup: `attacks/2001.09679__00/output.md`

- Problem statement: `attacks/2001.09679__00/prompt.md`
- Source paper: Z. Dvořák, *A note on sublinear separators and expansion*, arXiv:2001.09679v2 (fetched via ar5iv).

B.1 Interpretation

Dvořák's actual definition (fetched verbatim from the paper): *"Let b_ε denote the supremum of real numbers b for which every hereditary class 𝒢 of graphs such that s_𝒢(n) = Θ(n^{1−ε}) satisfies ∇_𝒢(r) = Ω(r^b)."* His bounds are max(1/(2ε)−1, 0) ≤ b_ε (Corollary 2, via Plotkin–Rao–Smith, with a polylog loss) and b_ε ≤ 1/(2ε)−1/2 (Lemma 10, via ⌈m^{ε/(1−ε)}⌉-fold subdivisions of 3-regular expanders), plus b_ε = 0 for 1/2 ≤ ε ≤ 1 (Lemma 12). The concluding remarks ask whether the gap of 1/2 can be closed and raise possible discontinuity at ε = 1/2. The paper's conventions match the writeup's: balanced separator = every component of G−X has ≤ 2n/3 vertices; r-shallow minor = contraction of vertex-disjoint subgraphs of radius ≤ r; ∇ = maximum density |E|/|V| of an r-shallow minor.

The writeup instead formalizes b_ε as an **infimum**: inf{b : some hereditary class has s = Θ(n^{1−ε}) and ∇(r) = O(r^b)}. These two definitions are not a priori identical (a class with oscillating expansion could separate them). However, this does not matter for the verdict: the writeup's construction — a hereditary class C_ε with s_{C_ε}(n) = Θ(n^{1−ε}) and ∇_{C_ε}(r) = O(r^{1/(2ε)−1}) — implies, under **Dvořák's own** definition, that no b > 1/(2ε)−1 belongs to the sup-set (C_ε violates ∇ = Ω(r^b)), hence b_ε ≤ 1/(2ε)−1; combined with Dvořák's Corollary 2 (which puts every b < 1/(2ε)−1 in the sup-set, the polylog being absorbed by the strict inequality), this gives b_ε = 1/(2ε)−1 exactly. So the construction resolves the intended question — the upper bound improves to meet the lower bound — and also settles the secondary question by showing b_ε is *continuous* at ε = 1/2 (refuting the speculated discontinuity). A reasonable author of the original paper would consider the open remark resolved by this result, if correct. The missing one-paragraph bridge between the two definitions is the main reason this report is MINOR_GAPS rather than CONFIRMED.

The construction is genuinely different from Dvořák's (2-D grid blocks of L^2 vertices buying insulation distance L, with a global Euler-genus-O(g) cap on minor density, versus 1-D subdivided edges where distance L costs only L vertices); this is exactly the mechanism by which it beats his exponent by 1/2, and it is consistent with his "dimension 2 artifact" remark about b_{1/2} = 0.

B.2 Step-by-step findings

step	label	note
S1. Formalization of b_ε (Statement section)	GAP	Writeup uses inf-over-classes; Dvořák defines a sup of guaranteed lower-bound exponents. Not identical a priori, but the construction plus Dvořák's Corollary 2 yields b_ε = 1/(2ε)−1 under Dvořák's definition too (see Interpretation). Routine to repair; conclusion unaffected.

step	label	note
S2. Lemma 1(1): grid edge-isoperimetry $\lvert\partial A\rvert \geq c\cdot\mu(A)/L$	VALID	Standard (Cheeger constant $\Theta(1/\text{side})$ for the square grid); I re-proved it with an elementary full-rows/full-columns case analysis giving $c = 1$ for side m, and verified it exhaustively for all subsets of $m\times m$ grids, $m \leq 4$ (min of $\lvert\partial A\rvert\cdot m/\mu(A)$ was 2.0), and on large samples for $m = 5$, 6 (min 2.0–2.5).
S3. Lemma 1(2): port disagreements $D_i(A) \leq \lvert\partial_Q A\rvert + \mu(A)/L \leq C\lvert\partial_Q A\rvert$	VALID	The disjoint perpendicular-paths argument is correct (each path either contributes a boundary edge or L vertices to the minority side); degenerate case $\partial_Q A = 0$ forces $\mu = 0$ by (1). Verified exhaustively on small grids: the inequality $D \leq \lvert\partial A\rvert + \mu/\ell$ was never violated (max slack ≤ -2).
S4. Lemma 1(3): L disjoint separating crosscuts per port, in disjoint collars	VALID	Explicit nested U-paths of depth $k = 1..L$ work; side 40L vs port length L leaves the four collars (width ~3L, depth L, centred) far from corners. Verified computationally on the real block (side 121, $L = 3$): all 12 crosscuts pairwise disjoint and each separates its port from the other three ports.
S5. Existence of 4-regular α-edge-expanders H_g for all large g, α absolute (eq. 3)	VALID	Standard (random regular graphs, Bollobás 1988); uniformity in g is well known.
S6. Construction of X(g,L); $N = gM_L = \Theta(gL^2)$; bounded degree (eq. 4)	VALID	Verified computationally: $N = 87846 = 6\cdot121^2$ for $(g,L) = (6,3)$; max degree 4 (grid boundary vertices have degree 3 plus one matching edge).
S7. Euler genus O(g) of X(g,L) (eq. 5)	VALID	Ribbon-graph surface of H_g: g disks + 2g bands (4-regular $\Rightarrow$ $\lvert E(H_g)\rvert = 2g$), order-preserving matchings drawn without crossings inside the bands; Euler genus $\leq 2(\lvert E\rvert-\lvert V\rvert+1) = O(g)$.
S8. Lemma 2: density of genus-γ graphs is $O(\sqrt{\gamma+1})$	VALID	Re-derived: $e \leq 3q-6+3\gamma$ from Euler's formula plus $e \leq q(q-1)/2$; case split at $q = \sqrt{\gamma+1}$ gives density $\leq 3+3\sqrt{\gamma}$ resp. $< \sqrt{\gamma}/2$.

step	label	note
S9. Eq. (7): every minor of X(g,L) has density O(√g)	VALID	Minors preserve embeddability in the same surface; apply S7 + S8.
S10. Lemma 3: $\lvert\partial_X S\rvert \geq (c/L)\cdot\min(\lvert S\rvert, N-\lvert S\rvert)$	VALID	Re-derived the whole chain: (9) is S2 per block; (10) holds because each non-crossing matching edge at an interface of $\delta_H(U)$ forces a state-disagreeing port vertex in one of the two blocks, bounded via S3 by $C(b_u+b_v)$; summing with deg(H) = 4 gives (11) with $C_1 = 1+4C$; (12) is the sum of (9); the final chain $\min(\lvert S\rvert,N-\lvert S\rvert) \leq M_L\cdot\min(\lvert U\rvert,g-\lvert U\rvert) + \Sigma\mu_v \leq (M_L/\alpha)\lvert\delta(U)\rvert + C_2L\lvert\partial_X S\rvert \leq C_3L\lvert\partial_X S\rvert$ checks out using $M_L = \Theta(L^2)$.
S11. Corollary 4: sep(X(g,L)) = Ω(gL)	VALID	The grouping step (a union A of components with $N/6 \leq \lvert A\rvert \leq N/2$ exists when $\lvert Z\rvert < N/6$ and all components $\leq 2N/3$) is correct in all three cases (one middle-size component; one component > N/2, take the rest; all small, greedy). $\lvert\partial_X A\rvert \leq \Delta\lvert Z\rvert$ with $\Delta \leq 5$; Lemma 3 finishes.
S12. Lemma 5: every subgraph F has a balanced separator O(n/L + √n)	VALID	Averaging over the L disjoint crosscuts gives $\lvert Z\rvert \leq 4n/L$; deleting them confines each component of F−Z to one block's interior/collar or to two matched collars plus their matching band (planar in all cases); planar separator on the unique big component (if any) adds O(√n) and yields balance (pieces $\leq (2/3)\lvert K\rvert \leq 2n/3$, other components total < n/3). Verified computationally on X(6,3): for F = X and for a random induced subgraph, every component of F−Z was planar and met ≤ 2 blocks (adjacent in H when 2), with $\lvert Z\rvert$ within the 4n/L bound.

step	label	note
S13. Eq. (16): sep(X(g,L)) = Θ(gL)	VALID	Upper bound from S12 with $n = \Theta(gL^2)$ (and also directly: removing all 4gL port vertices leaves components of size ≤ M_L; verified computationally: |Z| = 72 = 4gL, max component 14629 ≤ 2N/3 = 58564); lower bound is S11.
S14. Lemma 6: r-shallow minors with r < ηL have density O(1)	VALID	Block-locality claims re-derived (ports 30L apart ⟹ a radius-r set meets ≤ 2, necessarily H-adjacent, blocks; centers of J-adjacent branch sets are ≤ 2r+1 apart). The oriented-edge charging is correct: (W_v,E_v) is a minor of the bounded-genus subgraph on ≤ 17 blocks of N^2_{H}[v] (branch sets stay inside those blocks and remain connected there), so |E_v| ≤ C_0|W_v| by S8, and each minor vertex lies in ≤ 5 sets W_v, giving |E(J)| ≤ $5C_0$|V(J)|. Locality constants verified computationally on X(6,3): every ball of radius ≤ 59 ≈ 19.7L meets ≤ 2 blocks; min in-block port distance 118 ≥ 30L = 90; min distance to a non-H-adjacent block 120 = 40L.
S15. Eq. (19): ∇_r(X) ≤ C for r < ηL, ≤ C√g for all r	VALID	S14 plus S9; r = 0 case is the bounded-degree density ≤ 2.
S16. Parameter choice (20)–(22): p = 1/ε−2, g_L = Θ(L^p), N_L = Θ(L^{1/ε})	VALID	Arithmetic re-derived: p > 0 iff ε < 1/2; N_L = Θ(L^{p+2}) = Θ(L^{1/ε}). 4-regular expanders exist for every large g, so g_L can track L^p with N_{L+1}/N_L → 1.
S17. Eq. (23): sep(X_L) = Θ(N_L^{1−ε})	VALID	(p+1)/(p+2) = (1/ε−1)/(1/ε) = 1−ε.

step	label	note
S18. Eqs. (24)–(25): $s_{C_\varepsilon}(n) = \Theta(n^{1-\varepsilon})$, incl. exact-n convention	GAP	Upper bound is uniform and correct ($n \le N_L \Rightarrow n^\varepsilon = O(L) \Rightarrow n/L = O(n^{1-\varepsilon})$; $\sqrt{n} \le n^{1-\varepsilon}$ needs $\varepsilon < 1/2$, which holds). For the per-n (exact-order) reading of s_C, the isolated-vertex padding remark is correct but terse: the class as defined ("all subgraphs of all X_L") does not literally contain $X_L \sqcup I_k$, so C_ε must be enlarged to $\{F \sqcup I_k : F \subseteq X_L, k \ge 0\}$; this stays monotone, adds nothing to ∇, keeps the $O(n^{1-\varepsilon})$ upper bound, and the padded lower bound goes through ($k = N_{L+1}-N_L = o(N_L)$ since $N_{L+1}/N_L \to 1$, and the grouping argument of S11 survives the $o(N_L)$ perturbation). Routine repair; I checked the repaired argument in detail.
S19. Eq. (26): $\nabla_{C_\varepsilon}(r) = O(r^{1/(2\varepsilon)-1})$	VALID	Shallow minors of subgraphs of X_L are shallow minors of X_L; for $L > r/\eta$ density $O(1)$, for $L \le r/\eta$ density $O(\sqrt{g_L}) = O(L^{p/2}) = O(r^{p/2})$; constants uniform in L; $p/2 = 1/(2\varepsilon)-1$.
S20. Conclusion $b_\varepsilon = 1/(2\varepsilon)-1$ and continuity at $\varepsilon = 1/2$	VALID (modulo S1)	Follows under Dvořák's definition via the bridge described in the Interpretation section, using his Corollary 2 and Lemma 12; the writeup states the conclusion without performing the bridge.

No step was found to be an ERROR. The two GAP items (S1, S18) are presentational and repairable in a few lines each; neither threatens the result.

B.3 Reference check

- **arXiv:2001.09679 (Dvořák, *A note on sublinear separators and expansion*)** — fetched (ar5iv full text). Confirmed verbatim: the definition of b_ε (supremum form, hereditary classes, $s_\mathscr{G}(n) = \Theta(n^{1-\varepsilon})$, $\nabla_\mathscr{G}(r) = \Omega(r^b)$); Corollary 2 ($s_\mathscr{G}(n) = \Omega(n^{1-\varepsilon}) \Rightarrow \nabla_\mathscr{G}(r) = \Omega(r^{1/(2\varepsilon)-1}/\mathrm{polylog}\, r)$); Lemma 10 ($b_\varepsilon \le 1/(2\varepsilon)-1/2$, via subdivided 3-regular expanders); Lemma 12 ($b_\varepsilon = 0$ for $1/2 \le \varepsilon \le 1$); the concluding open remark about the gap of 1/2 and possible discontinuity at $\varepsilon = 1/2$; and the conventions (2n/3-balanced separators, radius-r shallow minors, density = $|E|/|V|$). The writeup's quoted lower bound $b_\varepsilon \ge 1/(2\varepsilon)-1$ matches Corollary 2 + definition.

- **Plotkin–Rao–Smith 1994 / Esperet–Raymond 2018** — invoked only through Dvořák's Corollary 2, which I take from the (refereed, published in EJC) source paper. Consistency check: PRS applied to the new class gives s = O(n^{1−ε}·polylog), compatible with the constructed s = Θ(n^{1−ε}); no contradiction (the construction sits exactly at the PRS boundary, optimal up to polylog).
- **Existence of 4-regular expanders with absolute edge expansion, uniformly in g** — standard (random regular graphs, Bollobás 1988); accepted.
- **Grid edge-isoperimetry, planar separator theorem, Euler's formula on surfaces** — standard; the first re-proved and machine-checked here (see below), the last re-derived in S8.
- **Priority**: Semantic Scholar lists only three papers citing arXiv:2001.09679 (sparse string graphs / linear expansion 2026; expansion of gap-planar graphs 2025; sketching distances in monotone classes 2022) — none addresses b_ε. Web searches for follow-ups closing the 1/2 gap found nothing, consistent with the catalog's own 2026-05 review. So the result does not appear to be already known, though absence of evidence in indexed literature is not proof of novelty.

B.4 Computational check

Scripts in `verification/scripts/2001.09679__00/` (Python 3.14, networkx 3.6.1):

1. `lemma1_grid_checks.py` — Lemma 1(1)–(2). Exhaustive over **all** subsets A of the m×m grid for m = 2, 3, 4 (up to 2^16 subsets) and 200k random + all rectangular subsets for m = 5, 6. Results: min over A of |∂A|·m/μ(A) was 2.00 (m=2), 3.00 (m=3), 2.00 (m=4), 2.50 (m=5, sampled), 2.00 (m=6, sampled) — the isoperimetric constant is comfortably absolute (worst case: half-grid). The port inequality D(A) ≤ |∂A| + μ(A)/ℓ was never violated (max slack ≤ −2 in every run). PASS.
2. `construction_checks.py` — builds X(g,L) exactly as specified with g = 6, L = 3 (blocks 121×121, N = 87846 vertices, 174276 edges) over a random 4-regular H. Results: (A) max degree 4 ≤ 5 — OK; (B) min in-block distance between distinct ports = 118 ≥ 30L = 90 — OK; (C) the 12 crosscuts are pairwise disjoint and each separates its port from the other three ports of its block — OK;

   (D) every ball of radius ≤ 59 (= 19.7L) meets at most 2 blocks, the nearest foreign block is always H-adjacent, and the nearest non-adjacent block is at distance 120 = 40L — this validates Lemma 6's locality with η ≈ 15–19, far more than the "sufficiently small η" needed; (E) removing all port vertices gives an explicit balanced separator of size 72 = 4gL with max component 14629 ≤ 2N/3 = 58564, confirming sep(X) = O(gL); (F) the Lemma 5 pipeline, run both on F = X and on a random induced subgraph (n = 52785), produced Z within the 4n/L bound with **every** component of F−Z planar (networkx planarity check) and meeting ≤ 2 (H-adjacent) blocks. PASS on every check.

No claimed property failed computationally. (The construction is an asymptotic family, so the code verifies the finite structural claims that the proofs rest on, not the asymptotic Θ/O statements themselves; those I re-derived by hand.)

B.5 Caveats

1. **Definition of b_ε (S1).** The writeup's inf-formalization is not Dvořák's sup-definition. The construction settles Dvořák's b_ε anyway, but the bridge (via his Corollary 2, absorbing the polylog) must be added; as written, the boxed "b_ε = 1/(2ε)−1" is proved for a nonequivalent-a-priori variant of b_ε.
2. **Exact-n separator convention (S18).** Dvořák's s_𝒢(n) reads per-n; the class must be formally enlarged with isolated-vertex paddings and the Ω(N_L/L) bound re-checked for the padded graphs. The writeup's one-sentence remark is correct in substance but does not note that the padded graphs are not subgraphs of any X_L.

3. **"Sufficiently large L / sufficiently small η."** Constants are never pinned down, but the computational check shows very generous slack (η up to ≈ 15 works against the needed "η small"; port distances 39L vs the claimed 30L).
4. **Expander existence for every g.** Needs 4-regular α-expanders for all large g with absolute α (true, standard) so that N_{L+1}/N_L ⟶ 1; the writeup asserts this via "the standard random regular graph argument" without citation.
5. **Edge cases.** ε near 0 (p large) and ε near 1/2 (p near 0, g_L growing slowly) both work since only $L \longrightarrow \infty$ asymptotics are used; $r = 0$ and disconnected subgraphs are unproblematic (bounded degree; Lemma 5 never assumes connectivity). The √n ≤ n^{1−ε} step genuinely requires $\varepsilon < 1/2$, exactly the announced range.
6. **Balancedness convention.** Both paper and writeup use 2n/3-balanced separators; no mismatch. Shallow-minor depth conventions also match (radius-r branch sets).
7. **Novelty** has been checked only against indexed/citing literature (3 citing papers, none relevant); a refereed-journal-grade priority search was not performed, matching the writeup's own caveat.

B.6 Referee summary

I came to this writeup expecting to find the usual single-pass failure mode and did not find one. The construction — blow each vertex of a bounded-degree expander into a $\Theta(L)\times\Theta(L)$ grid with four length-L boundary ports and join matched ports by order-preserving matchings — is genuinely different from Dvořák's subdivided expanders, and the mechanism by which it gains the extra factor of 1/2 in the exponent is sound: two-dimensional blocks buy insulation distance L at a cost of L^2 vertices (so small-radius shallow minors stay sparse, Lemma 6), while the whole graph embeds in a surface of Euler genus O(g), capping the density of *all* minors at O(√g) (Lemma 2). Every lemma survived detailed re-derivation: the block isoperimetry and port-disagreement bounds (verified exhaustively by machine on small grids), the global $\Theta(1/L)$-Cheeger bound for X(g,L), the $\Omega(gL)$ separator lower bound, the hereditary O(n/L + √n) separator via per-port crosscut averaging plus planarity of the resulting components (verified by machine on the real construction, including planarity of every component and the ≤ 2-blocks-per-component claim), the shallow-minor locality with generous constant slack (verified by machine), and the exponent arithmetic. The source paper was fetched and every quoted fact confirmed verbatim; no citing paper resolves the question. Two repairable defects keep this at MINOR_GAPS rather than CONFIRMED: the writeup silently replaces Dvořák's sup-based definition of b_ε with an inf-based one (the construction settles Dvořák's actual question, but the one-paragraph bridge through his Corollary 2 is missing), and the isolated-vertex padding needed for the per-n reading of s_𝒢(n) = Θ(n^{1−ε}) is sketched rather than executed (the class as defined does not contain the padded graphs). With those two paragraphs added, I believe this is a correct and publishable resolution of the open remark: b_ε = 1/(2ε)−1 for $0 < \varepsilon < 1/2$, and b_ε is continuous at $\varepsilon = 1/2$. Confidence is medium only because the argument is asymptotic and no machine check can certify the Θ/O statements themselves, and because an expert (Dvořák) stopping at 1/(2ε)−1/2 warrants residual humility; expert scrutiny of Lemmas 3, 5 and 6 is recommended before public claims.

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